

Table 2 shows the calculation.

Step 2: Find the residual sum of squares, written as Se or RSS.

The formulas are:

- y is the observed value
- $\hat{y}$ is the estimated value based on our regression equation (Note: The caret in $\hat{y}$ is affectionately called a hat, so we call this parameter estimate y -hat)
- $y - \hat{y}$ is called the residual and is written as e

The calculation is given in Table 3.

Note: How the formula derivation is arrived at is out of the scope of this article. *Hint:* Use calculus to differentiate $y = (B0 + B1x)^n$ with respect to x . Calculus is the mathematical study of continuous change, and differentiation cuts things into smaller pieces to find how they change.

Step 3: Find the regression equation using the B0 and B1 formulas.

- $B0 = \bar{y} - \bar{x} B1$
- $B1 = Sxy / Sxx$

Let's plug in the values we calculated in Step 1.

- $B1 = Sxy / Sxx = 3300.43/36.36 = 90.77$
- $B0 = \bar{y} - \bar{x} B1 = 333.57 - (3.34 * 90.77) = 30.40$

The regression equation is:

$$y = 30.40 + 90.77x$$

Note: The values above are rounded; hence, if you run through the program there will be slight variations. Also, here we should have normalised the value of the price of wine, to see non-biased results.

Here's a recap of what we have seen so far.

The relationship between the residuals and the slope B1 (alias a) and intercept B0 (alias b) is always as follows.

B1: Sum of products of x and y / sum of squares of $x = Sxy / Sxx$

- $B0$: Mean of y - mean of $x * B1 = \bar{y} - \bar{x} B1$

Cost function

This helps us find the best possible values of B0, B1, ..., BN, from where the average is taken by all the residuals. It is also known as the mean squared error (MSE) function.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Figure 6: MSE formula

$$RMSE = \sqrt{\frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2}$$

Figure 7: RMSE formula

We can also use the root mean squared error (RMSE) function as a cost function.

RMSE is more sensitive to outliers. But when the outliers are exponentially rare (like in a bell shaped curve), the RMSE performs very well and is generally preferred.

Correlation coefficient (R)

Now that we have the regression equation, let's see how accurate our predictions are when we use it. As per our expectation, shown in Figure 2, the line should be closest to the dots, i.e., the predicted values should be closer to the observed values (dots).

To help understand the accuracy of the predictions, we can use the correlation coefficient R to represent the accuracy of a regression equation. This is the same function used before to check the correlation of the data to make sure the linear regression model is a suitable candidate for this data.

The formulas are:

$R = \text{sum of products } y \text{ and } \bar{y} / \text{root of sum of squares of } y \text{ and sum of squares of } \bar{y}$

Data: $B1 = Sxy / Sxx =$

$3300.43/36.36 = 90.77$

$B0 = \bar{y} - \bar{x} B1 = 333.57 - (3.34 * 90.77) = 30.40$

The calculation is given in Table 4.

To check how much variance is explained by our regression equation, we use R^2 . This is also called the coefficient of determination.

Model training

It's really simple to code in Python, since

	Age of wine x	Price of wine y	Predicted price of wine $\hat{y} = B0 + B1x$ $\hat{y} = 30.40 + 90.77x$	Residuals ($e = y - \hat{y}$)	Squared residuals ($y - \hat{y}$) ²
1	0	75	30.11	44.89	2014.8
2	1.2	100	139.05	-39.05	1524.68
3	2.1	220	220.75	-0.75	0.56
4	3.4	300	338.76	-38.76	1502.24
5	4.1	440	402.3	37.7	1421.04
6	5.6	500	538.47	-38.47	1479.97
7	7	700	665.56	33.44	1186.15
Sum	23.4	2335			
Average	3.34 is $\bar{x}$	333.57 is $\bar{y}$			Se / 7 = 9129.42

Table 4